A-A-50804A
6 March 1986
SUPERSEDING A-A-50804
dated 17 April 1981

# COMMERCIAL ITEM DESCRIPTION

## VORTEX MIXER, LABORATORY, Single Cup, 115 volt, 50/60 Hz, AC

This commercial item description shall be used in preference to Military Specification MIL-S-36908A.

This commercial item description covers a variable speed laboratory vortex mixer for use in mixing tube contents of hand held tubes or small hand held flasks.

Salient characteristics:

Shall be a variable laboratory vortex mixer for use in mixing the fluid contents in hand held tubes or hand held small flasks.

Shall consist of the following:

1. A motor.
2. A metallic base (motor housing).
3. A cup receptacle.
4. A variable speed control.
5. An on-off switch.
6. An activation switch.
7. A line cord.

Shall be operable on 110 to 120 volt, 50-60 Hz AC supply lines. A 5/6 change in motor speed on a 50 Hz supply line from that on 60 Hz shall be acceptable.

Shall conform to the applicable requirements of Underwriters' Laboratories UL 1262 (Standard for Safety Laboratory Equipment) or UL 73 (Standard for Safety-Motor-Operated Appliances), with proof of conformance consisting of the following:

1. A UL label or UL listing of the product, provided a copy of the UL report is available to the procurement officer upon the latter's request.

2. A laboratory report from an independent laboratory, provided a copy is available to the procurement officer upon the latter's request. The report shall include actual test methods, data taken, and test results.

The cup receptacle shall be fabricated of neoprene, isoprene or polyurethane. The cup receptacle shall facilitate the mixing of fluid materials in test tubes from micro size up to and including 26 millimeters (mm) diameter x 200 mm long, and in other vessels of various sizes and shapes.

AMSC N/A FSC 6640

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The mixing action shall be the result of an eccentric motion of the motor driven receptacle cup. The mixing in a test tube shall be by manually positioning the test tube in a cavity in the top of the cup receptacle. The mixing in other vessels (e.g., flasks, beakers and odd shaped containers) shall be by manually holding such vessels against the side of the cup receptacle. The motion of the cup receptacle shall cause a whirlpool motion for the liquid contents of a test tube, and a turbulence for liquid contents of other vessels held against the side of the cup receptacle.

There shall be adequate clearance between the cup receptacle and base to preclude any contact during the operation of the laboratory mixer.

The on-off switch shall control the operation of the laboratory mixer for either continuous or intermittent mixing. The "on" position shall permit a continuous motor operation. With the switch in the "off" position, the motor operation shall be controlled by a pressure activation switch which shall be in the form of either:

1. An internally mounted switch that is actuated by external pressure being applied to the cup receptacle and then deactivated by the release of the external pressure.

2. A hand operated extended touch plate.

The variable speed control shall be in the form of a multi-position switch for stepless changes in speed.

Two sponge strips or four rubber feet shall be securely attached to the underside of the base. The exterior housing of the base shall have a suitable finish that is resistant to corrosion or scratching.

The line cord shall be three conductor (with ground lead) and shall terminate with a three blade male plug for use in receptacles specified in American National Standards Institute (ANSI) C73.11.

During operation, the laboratory mixer shall operate without any type of failure, and shall show no tendency to creep.

Workmanship. The laboratory vortex mixer shall be free from defects which detract from its appearance or impairs its serviceability.

Unit. Each (EA). One vortex mixer, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.



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Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspections. Inspections, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Tests. Tests shall be conducted to determine compliance with specification requirements. Each vortex laboratory mixer shall be tested to demonstrate its capability to operate properly.

Sampling for examination. Sampling for examination shall be conducted in accordance with commercial practice.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U. S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.



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Custodians:

Army-MD
Navy-MS
Air Force-03

CIVIL ENGINEERING COORDINATING ACTIVITY:

VA-OSS

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DH

Project No. 6640-1335

Location: 53-7-16

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| A-A-50804A | VORTEX MIXER, LABORATORY, Single Cup |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)* _____ |

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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